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An underwater acoustic semantic communication approach to underwater image transmission

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Abstract

We develop an underwater acoustic semantic communication approach for underwater image transmission in this study. The developed approach has two main merits. First, it enables efficient underwater image transmission, in terms of microvolume textual semantics, via the underwater acoustic channel, which typically has limited bandwidth. Second, it enhances robust underwater image transmission by leveraging semantic communication in the underwater environment, where the signal-to-noise ratio (SNR) is typically low. To achieve these goals, semantic representations for underwater images are developed. Specifically, at the transmitter's end, semantic extraction is conducted by converting an original underwater image into descriptive text, from which an entity list is subsequently identified. The entity list represents the textual semantics of the underwater image, providing a microvolume representation for efficient underwater image transmission. The entity list is then transmitted through the underwater acoustic channel to the receiver. At the receiver's end, semantic reconstruction is conducted by converting the received entity list back into descriptive text, which is subsequently converted into a generated underwater image semantically consistent with the original. The semantic extraction and semantic reconstruction of the underwater image are empowered by deep learning technologies with the support of a reliable semantic knowledge base. The experimental results prove that our approach offers advantages, in terms of microvolume and robust representations, over standard approaches to underwater image transmission. Therefore, this approach establishes a new baseline for underwater image transmission, which could potentially overcome the challenges posed by the narrow bandwidth and low SNR of the underwater acoustic channel.

Keywords Underwater image transmission, Semantic communication, Underwater acoustic channel, Underwater image captioning, Underwater image generation

1 Introduction

Underwater images play a crucial role in various fields of marine science research, such as underwater target detection (Jian et al. 2024) and underwater ecological monitoring (Du et al. 2016). Furthermore, the wireless

transmission of underwater images facilitates the communication between the underwater sensing unit and an observation point (which may be on board or on land), thus playing a vital role in underwater exploration.

Underwater image wireless transmission is typically enabled by underwater acoustic communication because sound is the primary means of long-distance wireless transmission in underwater environments. Underwater images generally have large volumes and high variability, requiring a communication channel with wide bandwidth and high signal-to-noise ratio (SNR) for efficient and robust transmission. However, an underwater acoustic

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channel generally exhibits narrow bandwidth and low SNR, posing unique challenges for underwater wireless transmission (Niu et al. 2023). High-frequency underwater sound is easily absorbed by water and is severely attenuated, which can complicate information transmission; only the low-frequency range remains relatively stable and conducive to effective transmission (Hildebrand 2009; Jia and Huang 2014). Consequently, the available bandwidth for the underwater acoustic channel is limited, hindering the efficient transmission of large-volume data. In addition, the underwater acoustic channel suffers from low SNR due to complex environmental noises, such as tides, waves, biological groups, and human activities in underwater environments (Kilfoyle and Baggeroer 2000; van Walree 2013), which significantly impair the reliability of data transmission (Chen and Xu 2020). Given these characteristics, underwater image wireless transmission through underwater acoustic communication faces significant challenges: limited bandwidth hinders transmission efficiency, and environmental noises impact transmission reliability. Therefore, achieving effective underwater image transmission via wireless communication in underwater environments remains a longstanding challenge in the literature (Chang et al. 2019; Wang et al. 2019; Zhou et al. 2023).

Image wireless transmission using semantic communication has seen rapid advancements in recent years, showcasing its potential to significantly enhance the image transmission performance (Wheeler and Natarajan 2023). Semantic communication, an emerging intelligent technology, focuses on conveying the semantic meaning of data rather than the data itself (Xie et al. 2021; Luo et al. 2022). This technology compresses large-volume original data into microvolume semantics, reducing the bandwidth demands and enabling efficient data transmission (Zhang et al. 2022a). In addition, semantic communication utilizes deep learning technologies and the semantic knowledge base to reconstruct information, enhancing robustness against interference from complex environmental noises and ensuring reliable data transmission (Zhang et al. 2022b). Considering these advantages, we believe semantic communication is a valuable tool for enhancing the performance of underwater acoustic communication in underwater image wireless transmission scenarios (Ren et al. 2024). Therefore, we propose leveraging semantic communication to revolutionize underwater acoustic communication for effective underwater image transmission.

Existing research on underwater acoustic communication has focused on enhancing underwater image transmission performance through technological and algorithmic improvements. Cai et al. (2016) developed an algorithm combining compressive sensing and the

wavelet transform, efficiently capturing frequency information and reconstructing images at lower sampling rates. Cui et al. (2021) proposed a nonuniform, non-orthogonal multicarrier modulation scheme based on the frequency characteristics of compressed images, enhancing spectral efficiency and transmission reliability. With the widespread adoption of deep learning in image processing, optimizations in key processes, such as compression, have significantly enhanced the transmission performance. Su et al. (2021) developed an efficient image transmission algorithm based on reinforcement learning, which enhanced the image quality and reduced bit error rates in sea trials. Fang et al. (2023) introduced a compression framework based on convolutional neural networks (CNNs), leveraging underwater prior knowledge to achieve superior performance in both machine and human vision at extremely low bit rates. However, the narrow bandwidth and low SNR of the underwater acoustic channel significantly impair the efficiency and reliability of underwater image wireless transmission. Both standard underwater acoustic communication technologies and those leveraging deep learning have gradually reached the theoretical limits of channel capacity, thereby constraining further advancements in transmission performance (Niu et al. 2021). As a result, there is a pressing need for innovative breakthroughs in underwater image transmission using underwater acoustic communication.

Semantic communication extracts semantic features from original images at the transmitter's end and reconstructs them at the receiver's end, achieving high-quality image transmission. Huang et al. (2021) designed an image semantic encoding model from coarse to fine using generative adversarial networks (GANs) (Goodfellow et al. 2020). The model generates semantic features in the base layer and recovers image details in the enhancement layer, resulting in visually pleasing reconstructed images at low bit rates. Dong et al. (2023) proposed a layered semantic communication system for image transmission, achieving extraction, transmission, and reconstruction under varying semantic precision and channel conditions. Li et al. (2023) developed a dual-path semantic communication system operating at both the image and concept levels, improving the robustness of medical image transmission. Lokumarambage et al. (2023) used semantically segmented images for transmission and employed a pretrained GAN to reconstruct expected images, significantly reducing bandwidth demands. Fu et al. (2023) employed CNNs to extract multiscale semantic features and quantify them, enhancing the image reconstruction quality through the GAN-based training. While semantic communication offers substantial benefits for image wireless transmission, current

research primarily focuses on electromagnetic wave media, with seldom exploration of other media. Further research is necessary to expand into new domains, particularly underwater acoustic scenarios.

To address the challenges of underwater image transmission posed by the underwater acoustic channel and to expand semantic communication into this new domain, we propose a novel approach to underwater image transmission based on underwater acoustic semantic communication. The merits of the proposed approach are twofold.

First, the microvolume representation inherent in semantic communication effectively addresses the limited bandwidth of the underwater acoustic channel. By transmitting textual semantics rather than original underwater images, the data volume for transmission is reduced, and the demand for channel bandwidth is alleviated.

Second, we exploit the robustness of semantic communication to ensure reliable underwater image transmission even under low SNR conditions. By leveraging deep learning technologies with the support of a reliable semantic knowledge base, the robust reconstruction of underwater images is enhanced.

Our work expands semantic communication principles to underwater acoustic communication, promising

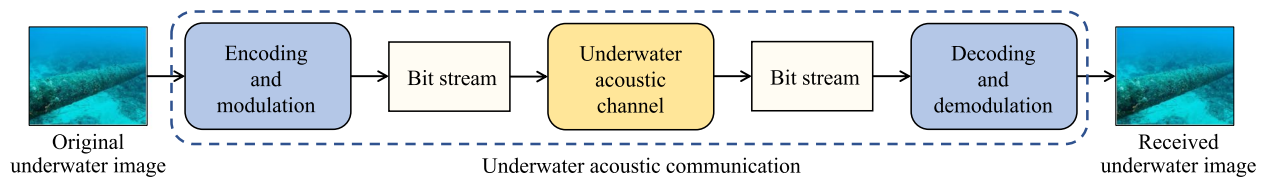
efficient and reliable underwater image transmission in challenging underwater environments.

The remainder of this paper is structured as follows. Section 2 presents the potential revolutionary shift from traditional underwater acoustic communication to the proposed underwater acoustic semantic communication for transmitting underwater images. Section 3 describes the specific implementation processes of underwater image transmission through underwater acoustic semantic communication, including semantic extraction, semantic reconstruction, and training processes. Section 4 presents comparative experiments with different underwater image transmission approaches and provides systematic analyses of the experimental results. Section 5 concludes the paper.

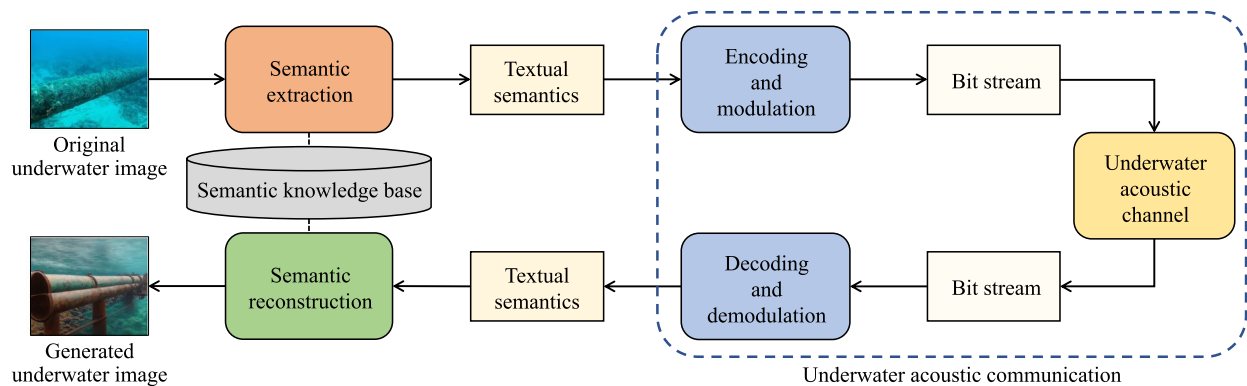
2 Revolutionary shift from underwater acoustic communication to underwater acoustic semantic communication

In this section, we discuss the revolutionary shift from traditional underwater acoustic communication to the proposed underwater acoustic semantic communication for transmitting underwater images. Figure 1 illustrates the two schemes for underwater image transmission.

Figure 1a illustrates the traditional scheme, where an original underwater image is encoded and modulated into a bit stream at the transmitter's end. The bit stream



(a) Underwater image transmission based on traditional underwater acoustic communication



(b) Underwater image transmission based on underwater acoustic semantic communication

Fig. 1 Revolutionary shift from underwater acoustic communication to underwater acoustic semantic communication for underwater image transmission

is transmitted through the underwater acoustic channel and subsequently decoded and demodulated at the receiver's end. Ultimately, it is reconstructed into the received underwater image. It is observed that the original underwater image is typically large in volume and demands substantial bandwidth for efficient transmission, which contrasts with the narrow bandwidth of the underwater acoustic channel. In addition, despite various channel coding schemes designed to enhance transmission reliability, the underwater acoustic channel's consistently low SNR still potentially impairs the transmission quality.

Figure 1b illustrates the underwater image transmission based on underwater acoustic semantic communication. Although the process of underwater acoustic communication remains similar to the traditional scheme, this scheme includes new modules, i.e., semantic extraction and semantic reconstruction. At the transmitter's end, the semantic extraction module converts the original underwater image into textual semantics. Then, they are encoded, modulated, and transmitted through the underwater acoustic channel to the receiver. At the receiver's end, the received bit stream is decoded and demodulated into the received textual semantics. They are then reconstructed into the generated underwater image based on the semantic reconstruction module. Notably, the volume of textual semantics is much smaller than that of the original underwater image, rendering microvolume representation for efficient transmission. In addition, both the semantic extraction and semantic reconstruction modules are empowered by deep learning technologies with the support of a reliable semantic knowledge base, thereby enhancing the reliability of underwater image transmission even in low SNR underwater environments. The generated underwater image may not match the original underwater image at a pixel-for-pixel level, but it maintains semantic consistency with the original. Therefore, underwater acoustic semantic communication has the potential to revolutionize underwater image transmission in terms of efficiency and reliability.

Table 1 compares underwater image transmission approaches between traditional underwater acoustic communication and the newly proposed underwater acoustic semantic communication. It is observed that the

new semantic communication approach offers significant advantages, providing a novel strategy for achieving high-quality underwater image transmission.

3 Implementation of underwater image transmission based on underwater acoustic semantic communication

In this section, we describe the implementation processes of underwater image transmission through underwater acoustic semantic communication, including semantic extraction, semantic reconstruction, and training processes.

3.1 Semantic extraction

The semantic extraction module aims to generate an entity list that represents the textual semantics of an underwater image. Figure 2 illustrates the specific implementation process of semantic extraction, which includes image captioning and entity identification submodules. The image captioning submodule converts the original underwater image into descriptive text, and the entity identification submodule processes the descriptive text into the entity list. Both submodules are detailed as follows.

3.1.1 Image captioning

The image captioning submodule generates descriptive text for an underwater image. Figure 3 illustrates the specific implementation process of this submodule, which includes an image encoder and a text decoder. The image encoder employs multiple convolutional networks to encode the original underwater image I into image features $\mathbf{x}$. The text decoder utilizes a long short-term memory (LSTM) network with an attention mechanism to decode these image features $\mathbf{x}$ into the descriptive text T .

3.1.2 Entity identification

The entity identification submodule converts descriptive text into an entity list. Figure 4 illustrates the specific implementation process of this submodule, which includes entity extraction and entity categorization. Entity extraction is performed using a BERT-BiLSTM-CRF model (Dai et al. 2019) to identify entities E from the descriptive text T . The bidirectional encoder

Table 1 Comparison between underwater acoustic communication and underwater acoustic semantic communication for underwater image transmission

Aspect	Transmission	Underwater acoustic communication	Underwater acoustic semantic communication
Source	Underwater images	Original underwater image data	Textual semantics extracted from underwater images
Efficiency	Narrow bandwidth	Bandwidth overwhelmed by large-volume data	Efficient bandwidth utilization with microvolume textual semantics
Reliability	Low SNR	Susceptible to noise and interference	High robustness through semantic representations

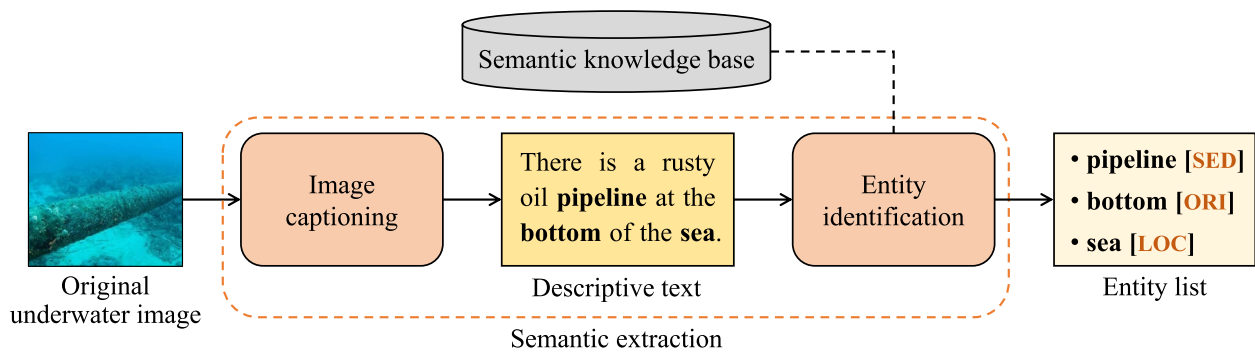


Fig. 2 Implementation of semantic extraction

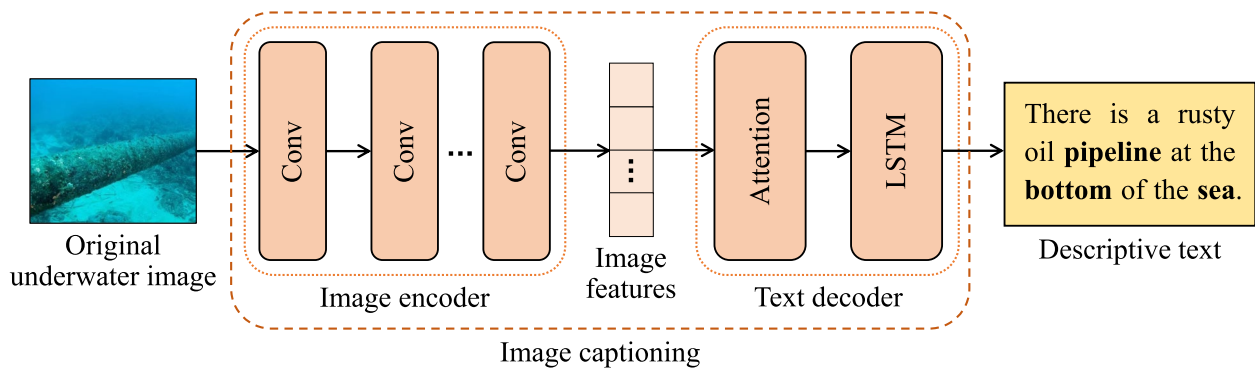


Fig. 3 Implementation of image captioning

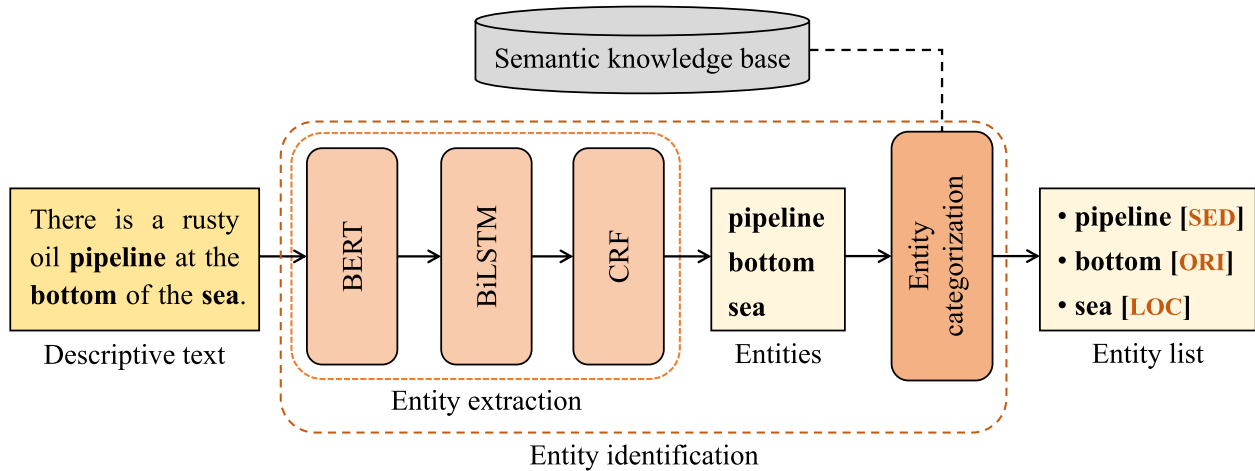


Fig. 4 Implementation of entity identification

representation of transformer (BERT) model (Devlin et al. 2019) processes the word sequence of T into a word embedding sequence that incorporates contextual information. The word embedding sequence is then input into a bidirectional long short-term memory (BiLSTM) network, which generates a feature vector sequence. The

conditional random field (CRF) layer selects the optimal entity indication sequence from all possible entity indication sequences based on the feature vector sequence. An entity indication sequence is a series of indicators that specify whether each word in the descriptive text is an entity or part of one. Finally, words or terms are

extracted as entities E from the descriptive text T . Entity categorization assigns each entity in E a category label, resulting in an entity list R , where each entry includes the entity and its category label. For example, an entry might appear as 'pipeline [SED]'; where 'pipeline' is the entity and 'SED' is the category label. In addition, semantic information, such as entities and their category labels, is integral to a semantic knowledge base, which is shared by both the semantic extraction and semantic reconstruction modules. The semantic knowledge base assists in resolving semantic ambiguities during transmission.

Observations: By simplifying underwater images into entity lists, semantic extraction significantly reduces the data volume for underwater image transmission. This process effectively addresses the narrow bandwidth of the underwater acoustic channel by transmitting textual semantics rather than full underwater images, thus reducing the data volume and easing the burden on channel bandwidth for more efficient transmission.

3.2 Semantic reconstruction

The semantic reconstruction module reconstructs a received entity list into an underwater image, which is supposed to be semantically consistent with the original underwater image. Figure 5 illustrates the specific implementation process of semantic reconstruction, which includes the word correction submodule, the text generation submodule, and the text-to-image submodule. The word correction submodule corrects any incorrect entities (i.e. words) in the received entity list and generates a corrected entity list. The text generation submodule then converts this corrected entity list into descriptive text. The text-to-image submodule converts the descriptive text into the generated underwater image. All three submodules are detailed below.

3.2.1 Word correction

The word correction submodule corrects any possible incorrect entities (i.e. words) in a received entity list by referencing a semantic knowledge base, ensuring reliable

information transmission. Each entity is evaluated to determine if its category label matches the category of the entity. A match confirms the correctness, whereas a mismatch requires modification. Corrected entities are generated for the incorrect entities based on their category labels and the semantic knowledge base, ensuring the reliability of the information. Given the received entity list $\mathcal{E}$ with possible incorrectness, the word correction submodule results in a corrected entity list $\mathcal{R}$. The category labels are not attached to the entities in the corrected entity list illustrated in Fig. 5, as these category labels are no longer required for subsequent procedures.

3.2.2 Text generation

The text generation submodule converts an entity list into descriptive text. Figure 6 illustrates the specific implementation process of this submodule, which includes draft construction and a text generator. The construction of the draft is implemented by converting the entity list $\mathcal{R}$ into a more comprehensible draft D . '<S>' is a mask symbol that represents the text content to be generated. The text generator is implemented on the basis of a bidirectional and autoregressive transformers (BART) model (Lewis et al. 2020), which takes the draft D as input and employs a bidirectional encoder and an autoregressive decoder to generate the descriptive text H . The bidirectional encoder uses a BERT-like approach to better understand the contextual information of the text. Meanwhile, the autoregressive decoder uses a transformer-like generative decoding approach, effectively generating the text content.

3.2.3 Text-to-image

The text-to-image submodule converts descriptive text into an underwater image. Figure 7 illustrates the specific implementation process of this submodule, which includes a contrastive language-image pretraining (CLIP) (Radford et al. 2021) text encoder and an image generator. The submodule utilizes a stable diffusion model (Rombach et al. 2022) as its core. The CLIP text encoder

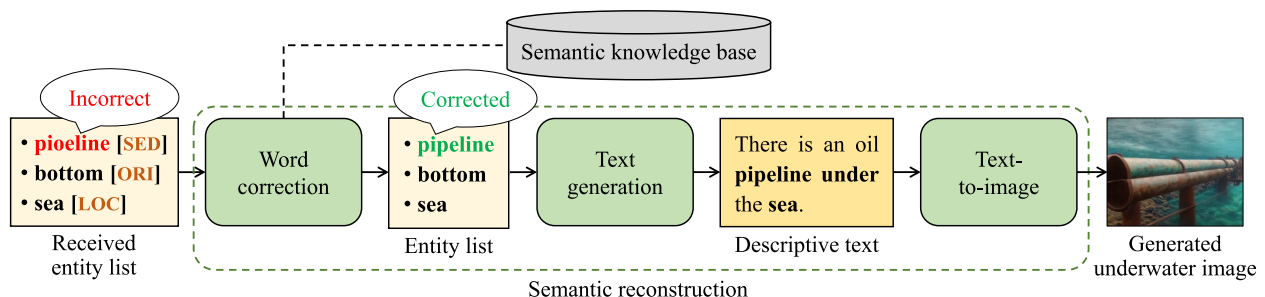


Fig. 5 Implementation of semantic reconstruction

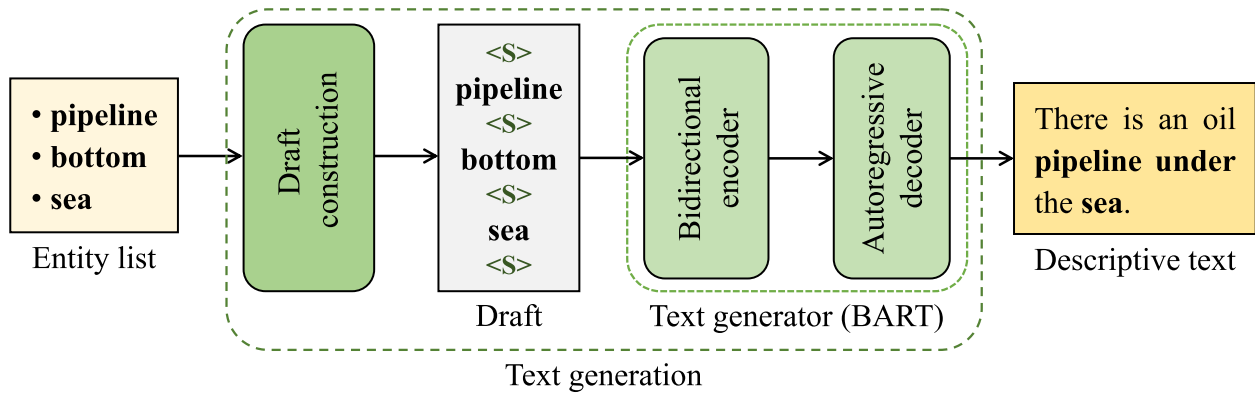


Fig. 6 Implementation of text generation

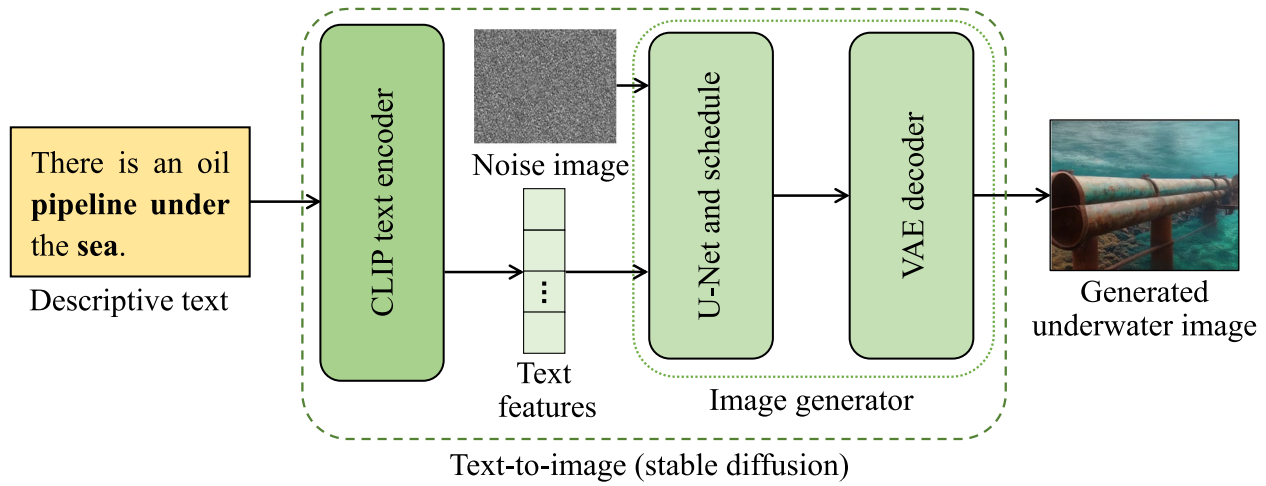


Fig. 7 Implementation of text-to-image

encodes the descriptive text H to generate the text features y . Within the image generator, a random noise image is initialized, and both the text features y and the random noise image are input into a U-Net (Ronneberger et al. 2015) network. The U-Net network iteratively denoises and generates image features with the scheduling algorithm for dynamical adjustment. A variational auto-encoder (VAE) (Kingma and Welling 2013) decoder processes these image features and converts them into a pixel-level generated underwater image $\mathcal{I}$.

Observations: Semantic reconstruction enhances the transmission reliability, particularly under low SNR conditions. It leverages deep learning technologies and a reliable semantic knowledge base to ensure the accurate reconstruction of underwater images even in challenging environmental conditions. Although the generated underwater image may not achieve the pixel-level reconstruction of the original, it preserves the semantic consistency. This new underwater image transmission

strategy is normally adequate for applications where the semantic contents of underwater images are more crucial than their visual details.

3.3 Training processes

In this subsection, we describe the training processes for three components, i.e., the image captioning submodule in Fig. 3, entity extraction in Fig. 4, and the text generator in Fig. 6.

3.3.1 Image captioning training

We use an underwater image captioning dataset $\mathcal{D}_{IC} = \{I^{(a)}, T^{(a)}\}_{a=1}^A$ to train the image captioning submodule, where $I^{(a)}$ denotes an underwater image, $T^{(a)}$ denotes its corresponding descriptive text, i.e., the caption, and A denotes the number of samples for training. The dataset used in the study will be introduced in Section 4.1. During the training, the submodule takes the underwater image $I^{(a)}$ as input and generates the predicted

descriptive text $\hat{T}^{(a)}$ through the image encoder and the text decoder. To train the image captioning submodule, the disparity between the ground truth descriptive text $T^{(a)}$ and the predicted descriptive text $\hat{T}^{(a)}$ is minimized subject to the loss function $\mathcal{L}_{IC}$ as follows:

$$\mathcal{L}_{IC} = - \sum_{a=1}^A \sum_{k=1}^K \sum_{m=1}^M g_{m,k}^{(a)} \ln \hat{t}_{m,k}^{(a)}, \quad (1)$$

where $\hat{t}_{m,k}^{(a)}$ denotes the probability of the k -th vocabulary word being present as the m -th word in the predicted descriptive text $\hat{T}^{(a)}$, $g_{m,k}^{(a)}$ denotes an indicator that is 1 if the k -th vocabulary word is present as the m -th word in the ground truth descriptive text $T^{(a)}$ and 0 otherwise, M denotes the length of the predicted descriptive text $\hat{T}^{(a)}$, and K denotes the size of the vocabulary.

3.3.2 Entity extraction training

We use the dataset $\mathcal{D}_{EE} = \{T^{(b)}, L^{(b)}\}_{b=1}^B$ to train entity extraction, where $T^{(b)}$ denotes a descriptive text, $L^{(b)}$ denotes its corresponding entity indication sequence, and B denotes the number of samples for training. The dataset used in our study will be introduced in Section 4.1. For a descriptive text $T^{(b)} = [t_1^{(b)}, t_2^{(b)}, \dots, t_N^{(b)}]$, its n -th entry $t_n^{(b)}$ is the n -th word in the descriptive text $T^{(b)}$. For an entity indication sequence $L^{(b)} = [l_1^{(b)}, l_2^{(b)}, \dots, l_N^{(b)}]$, its n -th entry $l_n^{(b)}$ is the entity indicator for the n -th word in the descriptive text $T^{(b)}$. Specifically, $l_n^{(b)}$ specifies whether $t_n^{(b)}$ is an entity or part of one. The path score $s(T^{(b)}, L^{(b)})$ for the entity indication sequence $L^{(b)}$ is defined as follows:

$$s(T^{(b)}, L^{(b)}) = \sum_{n=1}^{N-1} F(t_n^{(b)}, l_{n+1}^{(b)}) + \sum_{n=1}^N Z(t_n^{(b)}, l_n^{(b)}), \quad (2)$$

where $F(t_n^{(b)}, l_{n+1}^{(b)})$ denotes the transfer score from the entity indicator $l_n^{(b)}$ to the entity indicator $l_{n+1}^{(b)}$, and $Z(t_n^{(b)}, l_n^{(b)})$ denotes the state score of word $t_n^{(b)}$ corresponding to entity indicator $l_n^{(b)}$.

During the training, we input the descriptive text $T^{(b)}$ into the BERT-BiLSTM-CRF model to generate the predicted entity indication sequence $\hat{L}^{(b)}$. The disparity between the ground truth entity indication sequence $L^{(b)}$ and the predicted entity indication sequence $\hat{L}^{(b)}$ is minimized subject to the loss function $\mathcal{L}_{EE}$ as follows:

$$\mathcal{L}_{EE} = - \sum_{b=1}^B \left(s(T^{(b)}, L^{(b)}) - \ln \sum_{\hat{L}^{(b)} \in \mathbb{L}^{(b)}} e^{s(T^{(b)}, \hat{L}^{(b)})} \right), \quad (3)$$

where $\mathbb{L}^{(b)}$ denotes all possible entity indication sequences.

3.3.3 Text generator training

We use the dataset $\mathcal{D}_{TG} = \{D^{(c)}, H^{(c)}\}_{c=1}^C$ to train the text generator, where $D^{(c)}$ denotes a draft, $H^{(c)}$ denotes its original descriptive text, and C denotes the number of samples for training. The dataset used in our study will be introduced in Section 4.1. During the training, we use a bidirectional encoder and an autoregressive decoder to convert the draft $D^{(c)}$ into the predicted descriptive text $\hat{H}^{(c)}$. Both the bidirectional encoder and the autoregressive decoder are initialized with the BART weights. To train the text generator, the disparity between the ground truth descriptive text $H^{(c)}$ and the predicted descriptive text $\hat{H}^{(c)}$ is minimized subject to the loss function $\mathcal{L}_{TG}$ as follows:

$$\mathcal{L}_{TG} = - \sum_{c=1}^C \sum_{j=1}^J \sum_{q=1}^Q \rho_{q,j}^{(c)} \ln \hat{h}_{q,j}^{(c)}, \quad (4)$$

where $\hat{h}_{q,j}^{(c)}$ denotes the probability of the q -th vocabulary word being present as the j -th word in the predicted descriptive text $\hat{H}^{(c)}$, $\rho_{q,j}^{(c)}$ denotes an indicator that is 1 if the q -th vocabulary word is present as the j -th word in the ground truth descriptive text $H^{(c)}$ and 0 otherwise, J denotes the length of the predicted descriptive text $\hat{H}^{(c)}$, and Q denotes the size of the vocabulary.

4 Experiments

4.1 Experimental setup

We use the underwater image captioning dataset (Li et al. 2025) as the baseline for our experiments. This dataset includes 3176 underwater images, each accompanied by five descriptive texts as five captions. Leveraging this dataset and its descriptive texts, we train the image captioning submodule, entity extraction, and the text generator. The detailed training hyperparameters are provided in Table 2. The experiments are conducted using the PyTorch deep learning framework on a system equipped with an Intel® Xeon® Gold 5218R CPU @ 2.10GHz (80 cores), 376 GB of RAM, and an NVIDIA GeForce RTX 2080Ti GPU. Further details on the training settings and

Table 2 Training hyperparameters for the image captioning submodule, entity extraction, and the text generator

Hyperparameter	Image captioning	Entity extraction	Text generator
Learning rate	10^{-4}	3×10^{-5}	5.6×10^{-5}
Epochs	160	32	64
Batch size	16	16	32
Dropout	0.5	0	0.1
Weight decay	0	0.01	0.01
Optimizer	Adam	AdamW	AdamW

experimental configurations are presented in the remaining parts of this subsection.

The image captioning submodule in Fig. 3 is trained with the underwater image captioning dataset, according to the training process detailed in Section 3.3.1. The dataset is divided into training, validation, and testing sets with a ratio of 6:2:2.

For entity extraction, we use descriptive texts from the underwater image captioning dataset and apply the begin-inside-outside annotation approach (Ramshaw and Marcus 1999) to label each word in the descriptive texts by an entity indicator. Table 3 illustrates an example of labeling words in a descriptive text by entity indicators, where ‘B-ENI’ indicates an entity and ‘O’ indicates a nonentity. This results in a dataset including 13438 descriptive texts with their corresponding entity indication sequences. Entity extraction in Fig. 4 is trained on the basis of this dataset, according to the training process detailed in Section 3.3.2. The dataset is divided into training, validation, and testing sets with a ratio of 8:1:1.

For the text generator, we use descriptive texts from the underwater image captioning dataset and generate drafts through a three-step process: extraction, projection, and masking. In the extraction step, we use the yet another keyword extractor algorithm (Campos et al. 2020) to identify keywords or key phrases, capturing the essential information from the descriptive texts. In the projection step, these keywords or key phrases are mapped back to the original descriptive texts. In the masking step, we apply extreme and selective masking rules, replacing other parts of the descriptive texts with mask symbols to generate drafts. This process results in a dataset of 12000 drafts with their original descriptive texts. The text generator in Fig. 6 is trained on the basis of this dataset, according to the training process detailed in Section 3.3.3.

In addition, entity categorization and the text-to-image submodule are implemented according to off-the-shelf schemes. Entity categorization in Fig. 4 operates by assigning a category label to each entity. We define nine category labels implying different category information. Table 4 presents the nine category labels and their corresponding category information, with an entity example given to each category label. According to Table 4, entity categorization results in an entity list for given

Table 4 Category labels for entities

Category label	Category information	Entity example
SED	Ecology: sediment	Pipeline
FIS	Ecology: fish	Butterfly fish
ANI	Ecology: animal	Sea turtle
PER	Ecology: person	Diver
COR	Ecology: coral	Coral reef
LOC	Space: location	Sea
ORI	Space: orientation	Bottom
HUE	Attribute: color	Red
NUM	Attribute: amount	Two

entities. The text generator in Fig. 6 is pretrained using the weights of the BART model (Lewis et al. 2022) to convert a draft into descriptive text. The text-to-image submodule in Fig. 7 uses the trained stable diffusion model (Rombach and Esser 2022) to convert descriptive text into an underwater image.

To validate the feasibility of underwater acoustic semantic communication for underwater image transmission, we conduct comparative experiments for the underwater acoustic semantic communication (UASC) approach and four standard image transmission approaches. Underwater acoustic communication is simulated according to the schemes in Fig. 1. In the experiments, the encoding and modulation module is simulated in terms of source encoding, convolutional encoding for channel encoding, and OFDM technology for modulation. The transmitting bit streams through the underwater acoustic channel are simulated in terms of an ArlPy underwater acoustic modeling toolbox (Chitre 2023) and a Bellhop acoustic propagation model (Porter 2011). The decoding and demodulation module is simulated in terms of OFDM technology for demodulation, convolutional decoding for channel decoding, and source decoding. The four comparison image transmission approaches use JPEG, JPEG2000, WebP, and better portable graphics (BPG) for source encoding/decoding. In contrast, the UASC approach uses textual semantics in terms of entity lists for source encoding/decoding. The four comparison image transmission approaches and the UASC approach are consistent in all the other settings.

Table 3 Entity extraction

Descriptive text	There	is	a	rusty	oil	pipeline	at	the	bottom	of	sea	.
Entity indication sequence (i.e., entity indicators)	O	O	O	O	O	B-ENI	O	O	B-ENI	O	B-ENI	O
Entity	–	–	–	–	–	Pipeline	–	–	Bottom	–	Sea	–

4.2 Evaluation metrics

4.2.1 Image quality score

We evaluate image quality using several nonreference metrics: blind/referenceless image spatial quality evaluator (BRISQUE) (Mittal et al. 2012), natural image quality evaluator (NIQE) (Mittal et al. 2013), perception-based image quality evaluator (PIQE) (Venkatanath et al. 2015), hyper network-based image quality assessment (HyperIQA) (Su et al. 2020), multidimension attention network for image quality assessment (MANIQA) (Yang et al. 2022), and transformers for reference-free image quality assessment (TRES) (Golestaneh et al. 2022). BRISQUE, NIQE, and PIQE are traditional metrics that use statistical properties and perceptual models of images to evaluate quality. HyperIQA, MANIQA, and TRES leverage deep learning technologies to evaluate image quality through advanced perceptual feature extraction and modeling approaches. High scores for HyperIQA, MANIQA, and TRES indicate good image quality, whereas low scores for BRISQUE, NIQE, and PIQE indicate good image quality.

4.2.2 Semantic similarity score

We propose a semantic similarity score to quantify the degree of semantic similarity between the images. To compute the semantic similarity score of two images, they are first converted into descriptive texts using an image captioning model, and then, these descriptive texts

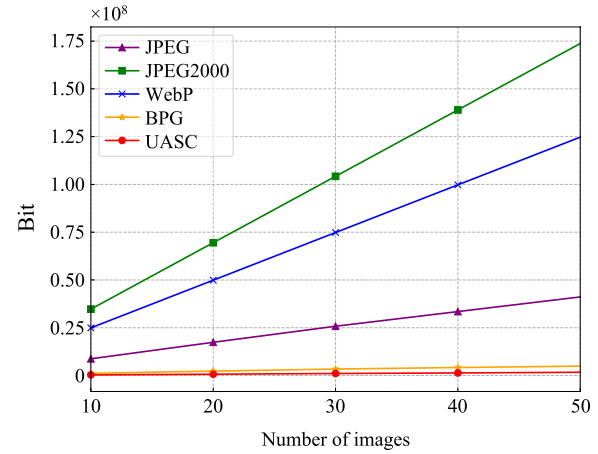


Fig. 8 Number of bits for transmission with respect to the number of transmitted images

are converted into two vectors using the BERT model. Subsequently, the cosine similarity score between the two vectors is computed to determine the semantic similarity score between the two images. To maintain consistency, we employ the trained image captioning submodule to convert the underwater images into descriptive texts, as illustrated in Fig. 3. The semantic similarity score $w(I, \mathcal{I})$ between the original image I and the generated image $\mathcal{I}$ is defined as follows:

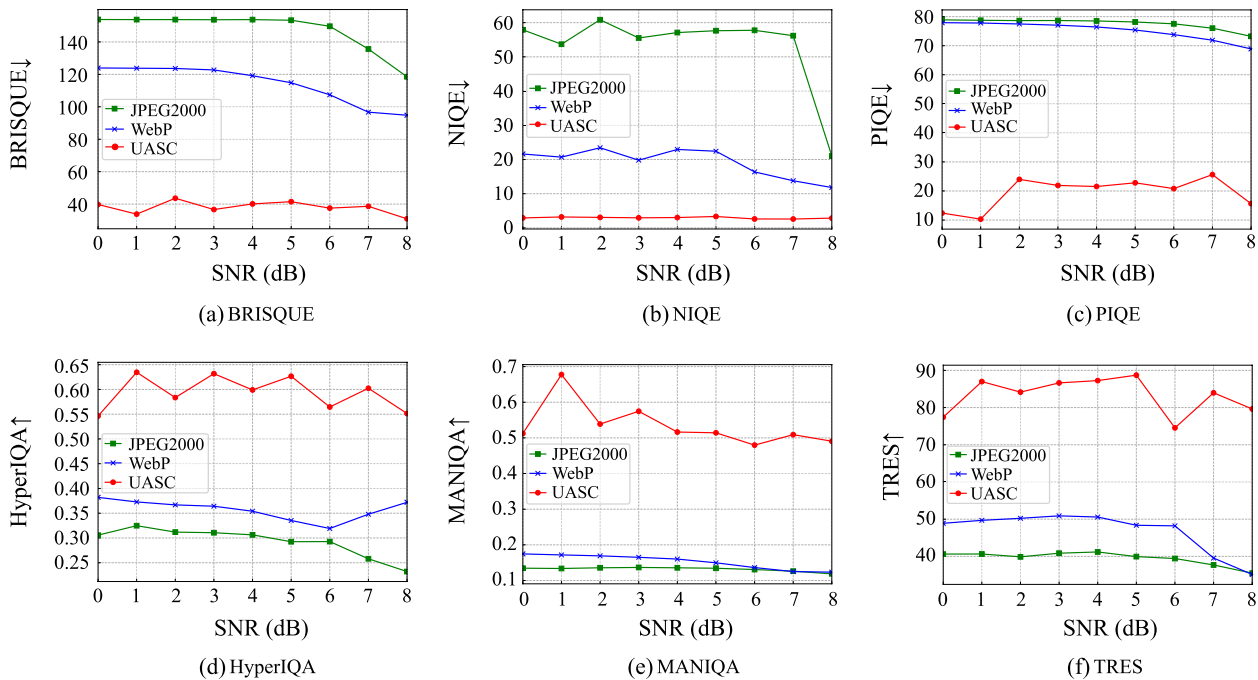


Fig. 9 Image quality scores for different image transmission approaches at varying SNR

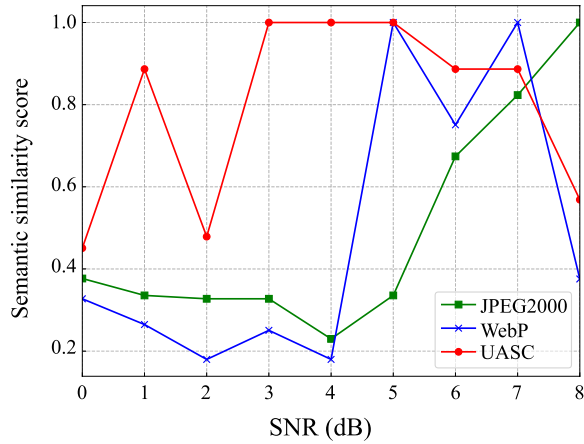


Fig. 10 Semantic similarity scores for different image transmission approaches at varying SNR

$$w(I, \mathcal{I}) = \frac{B_{\theta}(X_{\alpha}(I))(B_{\theta}(X_{\alpha}(\mathcal{I})))^T}{|B_{\theta}(X_{\alpha}(I))||B_{\theta}(X_{\alpha}(\mathcal{I}))|}, \quad (5)$$

where $X_{\alpha}(\cdot)$ represents the image captioning model and $B_{\theta}(\cdot)$ represents the BERT model. The semantic similarity score ranges from zero to one, with a score close to one indicating high semantic similarity.

4.3 Experimental analysis

4.3.1 Data volume for transmission

Figure 8 illustrates the number of bits for transmission with respect to the number of transmitted images. Among standard approaches, the BPG approach transmits the fewest bits due to its higher compression rate. Importantly, as the number of transmitted images

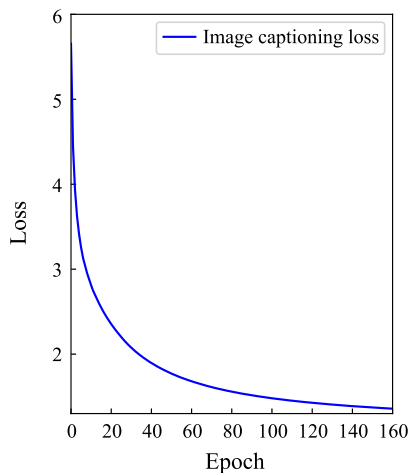
Table 5 P -values between the UASC approach and standard approaches under different evaluation metrics. SSC represents the semantic similarity score

	JPEG	JPEG2000	WebP	BPG
BRISQUE	None	0.004	0.004	None
NIQE	None	0.004	0.004	None
PIQE	None	0.004	0.004	None
HyperIQA	None	0.004	0.004	None
MANIQA	None	0.004	0.004	None
TRES	None	0.004	0.004	None
SSC	None	0.017	0.039	None

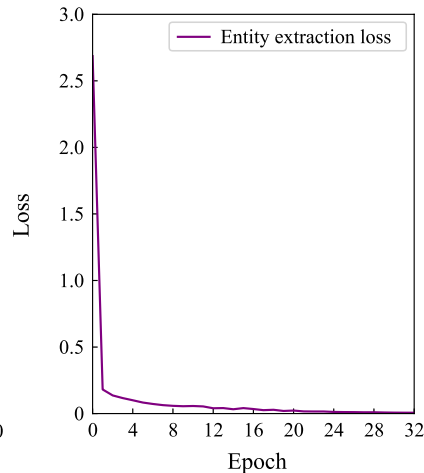
increases, the UASC approach consistently transmits fewer bits compared with standard approaches. This suggests a significant advantage in processing large volumes of underwater image data through semantic representations. In summary, the UASC approach reduces the data volume and the demand for channel bandwidth by transmitting textual semantics rather than underwater images. This underscores the benefits of microvolume representation through semantic communication for efficient underwater image transmission.

4.3.2 Image quality

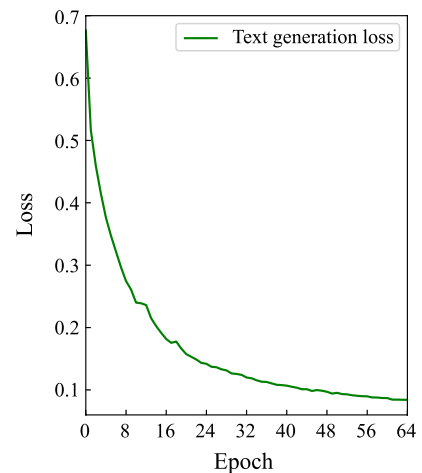
Figure 9 illustrates the image quality scores for different image transmission approaches at varying SNR. It is noteworthy that the underwater images generated by the JPEG and BPG approaches could not be obtained due to decoding errors within this SNR range.



(a) Image captioning loss



(b) Entity extraction loss



(c) Text generation loss

Fig. 11 Loss with respect to epoch

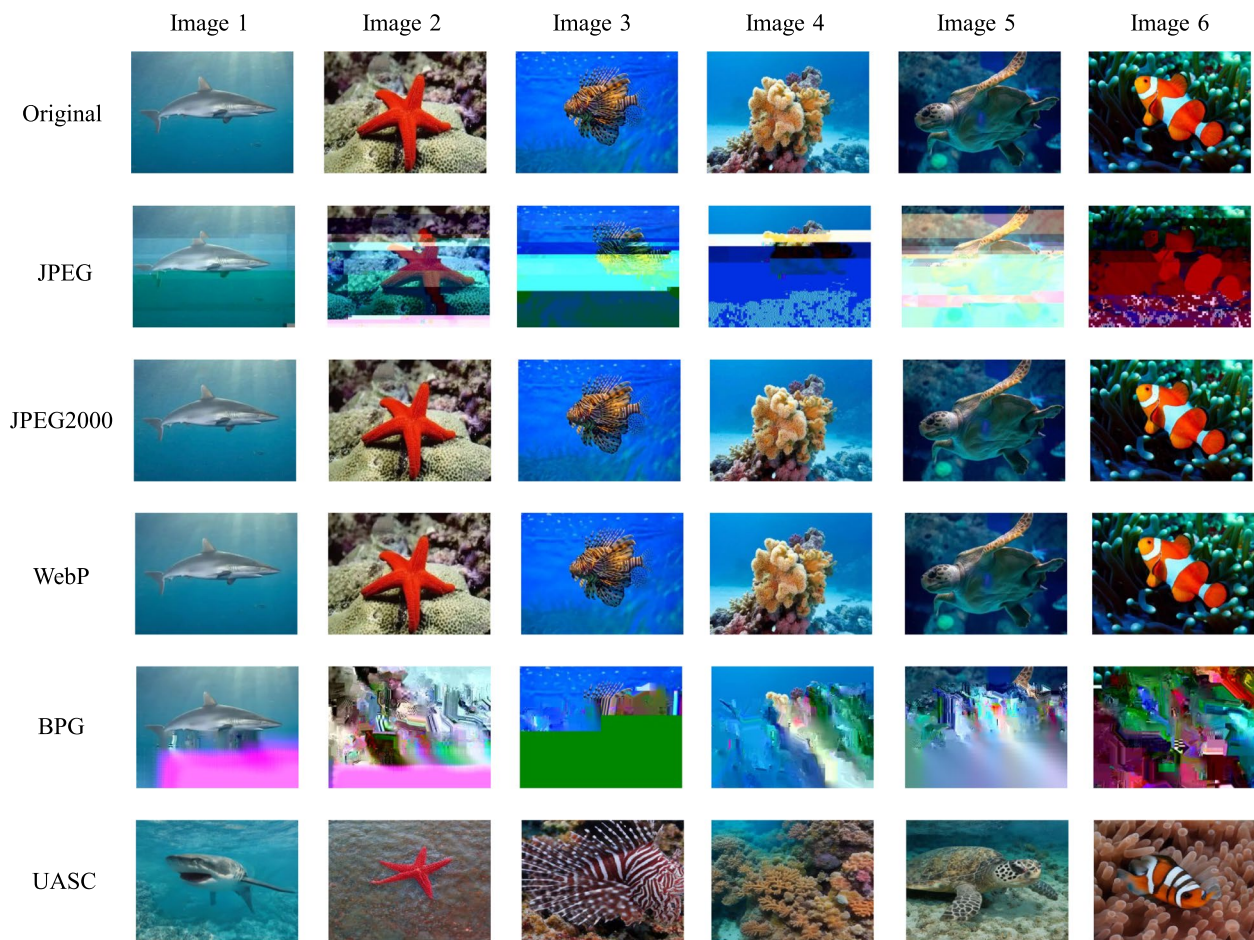


Fig. 12 Original underwater images and generated images from different image transmission approaches at SNR = 15

As illustrated in Figs. 9a–c, compared with standard approaches, the UASC approach consistently achieves low scores that remain stable. This suggests that, in traditional metrics, the visual quality of the underwater images generated by the UASC approach is superior, indicating significant advantages in terms of visual perception. Figures 9d–f illustrate that the UASC approach achieves high scores on the HyperIQA, MANIQA, and TRES metrics. This suggests that the underwater images generated by the UASC approach also exhibit superior visual perception under deep-learning-based evaluation metrics. In summary, the UASC approach generates high-quality underwater images under low SNR conditions and exhibits good robustness.

4.3.3 Semantic similarity

Figure 10 illustrates the semantic similarity scores for different image transmission approaches at varying SNR. Among standard approaches, the JPEG2000

approach showcases an overall trend in better scores, whereas the WebP approach reaches the highest score more quickly. Notably, at SNR conditions below 6 dB, the UASC approach consistently achieves high semantic similarity scores. This indicates that the UASC approach generates underwater images with high semantic similarity and demonstrates strong robustness under low SNR conditions. However, in a few scenarios, the UASC approach does not always achieve very high semantic similarity scores. This can be attributed to the nature of semantic communication, which prioritizes conveying the semantic meaning of underwater images over achieving pixel-perfect reproduction, thereby influencing these scores.

Both Figs. 9 and 10 underscore that the UASC approach exhibits enhanced robustness under low SNR conditions by leveraging deep learning technologies with the support of a reliable semantic knowledge base.

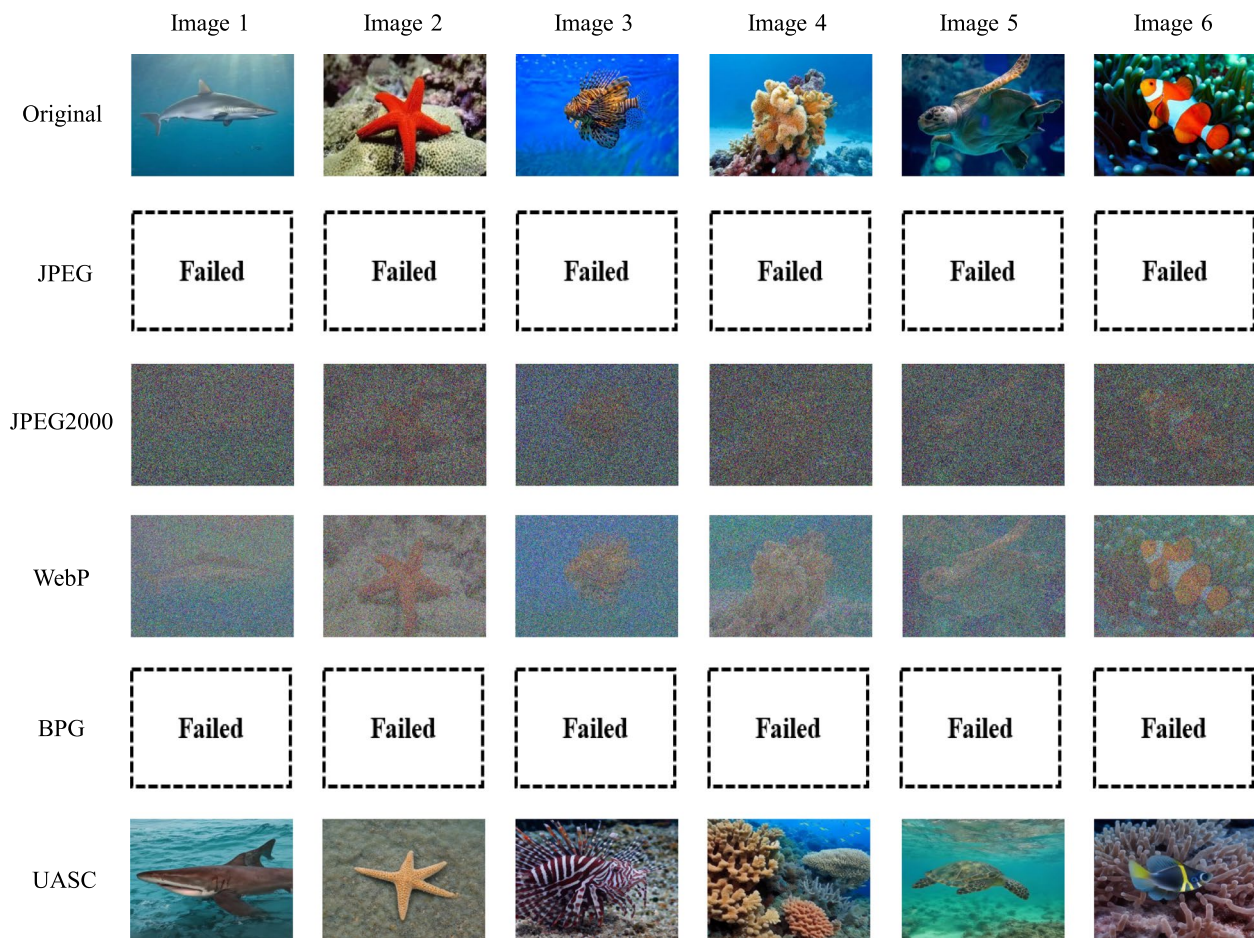


Fig. 13 Original underwater images and generated images from different image transmission approaches at SNR = 3

4.3.4 Statistical significance performance

To evaluate the statistical significance performance advantages of the UASC approach under different evaluation metrics, we use the Wilcoxon rank-sum test (Hollander et al. 2013) to compare the results of five random experiments between the UASC approach and the

standard approaches. When the P -value is less than 0.05, it indicates a significant performance difference. Table 5 illustrates the P -values between the UASC approach and the standard approaches under different evaluation metrics. The results confirm that the UASC approach

Table 6 Evaluation metric values for Image 3 and generated images from different image transmission approaches at SNR = 15

	JPEG	JPEG2000	WebP	BPG	UASC
BRISQUE↓	56.056	44.047	48.293	87.685	28.465
NIQE↓	4.009	4.337	4.713	8.012	3.367
PIQE↓	59.048	39.035	40.478	73.100	30.665
HyperIQA↑	0.454	0.583	0.628	0.413	0.627
MANIQA↑	0.210	0.524	0.537	0.158	0.674
TRES↑	63.619	82.781	88.325	62.927	74.819
SSC↑	0.215	0.999	0.999	0.329	0.944

Note: SSC represents the semantic similarity score, with the best scores highlighted in **bold**

Table 7 Evaluation metric values for Image 3 and generated images from different image transmission approaches at SNR = 3

	JPEG	JPEG2000	WebP	BPG	UASC
BRISQUE↓	None	126.824	117.988	None	30.983
NIQE↓	None	19.194	13.183	None	3.814
PIQE↓	None	74.959	71.983	None	34.055
HyperIQA↑	None	0.411	0.389	None	0.651
MANIQA↑	None	0.192	0.165	None	0.651
TRES↑	None	64.728	34.430	None	83.441
SSC↑	None	0.249	0.208	None	0.944

Note: SSC represents the semantic similarity score, with the best scores highlighted in **bold**

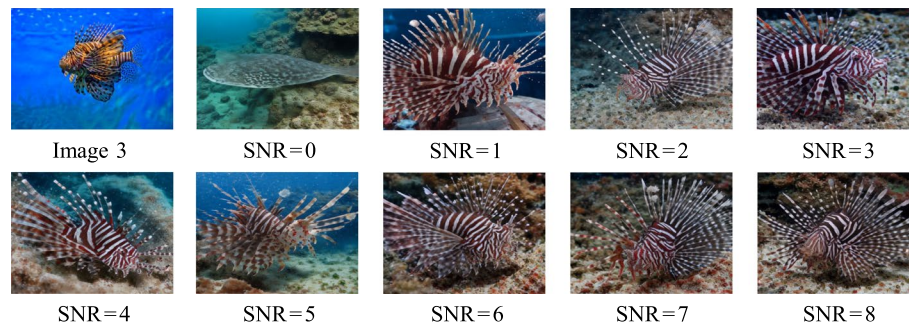


Fig. 14 Image 3 and its corresponding generated images from the UASC approach at varying SNR

demonstrates significant performance advantages compared to standard approaches under different evaluation metrics.

4.3.5 Convergence and stability analysis

Figure 11 illustrates the loss with respect to epoch for the image captioning submodule, entity extraction, and the text generator. Figures 11a–c are based on Eqs. (1), (3), and (4), respectively. They exhibit a gradual downward trend, reflecting that the models are effectively adjusting their parameters during training to better align with the data. In addition, the flattening of the loss curves toward the later stages of training suggests that the models are not encountering issues of overfitting or underfitting, further indicating strong generalization performance.

4.4 Experimental display

4.4.1 Different image transmission approaches at different SNRs

Figures 12 and 13 illustrate the original underwater images and the generated images from different image transmission approaches at SNRs of 15 and 3, respectively. Tables 6 and 7 illustrate the evaluation metric values for Image 3 and the generated images from different image transmission approaches at SNRs of 15 and 3,

respectively. The experimental results are discussed from both subjective and objective perspectives.

As illustrated in Fig. 12, standard approaches result in noticeable data loss with the JPEG and BPG approaches. The underwater images generated by the JPEG2000 and WebP approaches remain consistent with the original underwater images. Meanwhile, the UASC approach generates clear underwater images and maintains semantic consistency at high SNR. In addition, the UASC approach ensures consistency in appearance, such as color and species, as demonstrated by the underwater image generated from Image 2. As illustrated in Fig. 13, the standard approaches fail to generate underwater images using the JPEG and BPG approaches due to decoding errors. The underwater images generated by the JPEG2000 and WebP approaches exhibit varying levels of distortion, with the images generated by WebP exhibiting lighter distortion. In contrast, the UASC approach generates the clearest underwater images at low SNR, maintaining semantic consistency with the original underwater images.

As illustrated in Table 6, the UASC approach exhibits exceptionally well across most evaluation metrics. This indicates that the UASC approach maintains certain advantages at high SNR, ensuring that the generated underwater images align closely with human understanding. As illustrated in Table 7, the UASC

Table 8 Evaluation metric values for Image 3 and its corresponding generated images from the UASC approach at varying SNR

	SNR = 0	SNR = 1	SNR = 2	SNR = 3	SNR = 4	SNR = 5	SNR = 6	SNR = 7	SNR = 8
BRISQUE↓	35.573	38.799	38.683	30.983	33.188	37.506	32.635	34.929	31.530
NIQE↓	3.385	3.691	2.686	3.814	3.333	3.482	3.175	3.460	2.819
PIQE↓	32.708	35.818	25.290	33.964	27.146	32.816	32.400	33.045	24.333
HyperIQA↑	0.559	0.628	0.652	0.647	0.644	0.608	0.636	0.669	0.657
MANIQA↑	0.608	0.673	0.607	0.653	0.567	0.664	0.564	0.568	0.647
TRES↑	74.860	78.597	84.197	83.756	83.662	76.451	73.875	84.324	82.687
SSC↑	0.547	0.634	0.789	0.944	0.789	0.835	0.854	0.944	0.944

Note: SSC represents the semantic similarity score

approach demonstrates superior performance in the evaluation metrics. This implies that compared with standard approaches, the underwater images generated by the UASC approach achieve greater quality and higher semantic similarity at low SNR.

4.4.2 UASC approach at varying SNR

Figure 14 illustrates Image 3 and its corresponding generated images from the UASC approach at varying SNR. Table 8 illustrates the evaluation metric values for Image 3 and its corresponding generated images from the UASC approach at varying SNR. From a subjective view, the UASC approach generates clear and visually appealing underwater images, maintaining semantic consistency with Image 3 in most cases. From an objective view, the UASC approach demonstrates robust performance in all evaluation metrics, indicating its effectiveness in generating high-quality and high semantic similarity underwater images under low SNR conditions.

5 Conclusions

The proposed approach in this paper has leveraged semantic communication to significantly enhance the performance of underwater image transmission via underwater acoustic communication. By extracting the textual semantics from underwater images and representing them as entity lists, we have effectively alleviated the bandwidth limitations of the underwater acoustic channel. Through robust reconstruction using deep learning technologies and the semantic knowledge base, our approach ensures semantically consistent image reconstruction despite complex noises in underwater environments. Experimental results have validated that our approach surpasses standard approaches in the suitability for narrow bandwidths and low SNR conditions, confirming the practical feasibility of underwater acoustic semantic communication for underwater image transmission.

Moving forward, our future work will focus on deploying and refining the underwater acoustic semantic communication approach in real-world underwater environments. This will involve enhancing the adaptability of the approach to dynamic underwater environmental conditions, ensuring efficient and reliable data transmission.

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Additional information

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Authors' contributions

All authors have contributed to the conceptualization and design of the research. Ying Zhang and Peng Ren conceived the idea of this paper on underwater image transmission and wrote the manuscript. Huanyu Li provided the dataset and made critical modifications. Bingyu Li and Weibo Zhang conducted the literature research and data analysis. Li Li and Hao Wang made suggestive revisions and corrections to the manuscript.

Data availability

The data and references presented in this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Informed consent for publication was obtained from all participants.

Competing interests

The authors declare that they have no competing interests.

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